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Review

Machine Learning Applications in Fuel Reforming for Hydrogen Production in Marine Propulsion Systems

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Abstract

In the context of the shipping industry's transition towards low-carbon solutions, hydrogen energy exhibits substantial application potential in marine propulsion systems. Fuel reforming for hydrogen production represents one of the key technologies for efficient hydrogen production in maritime applications. Nevertheless, this process involves complex multi-scale reaction mechanisms, challenges in catalyst design, and difficulties in system optimization. This paper conducts a comprehensive review of the recent progress in the application of machine learning in fuel reforming hydrogen production technology. In the realm of catalysts, machine learning has expedited the design of efficient catalysts via high-throughput screening, performance prediction, and active site regulation. In reaction modeling, machine learning has facilitated the development of multi-scale kinetic models, enhancing the interpretability and predictive accuracy of reaction pathways. Regarding equipment and system optimization, machine learning has enabled innovations in reactor design, collaborative optimization of process parameters, and intelligent system control. This review aims to provide theoretical foundations and practical guidance for the technological development of ship propulsion systems. Moreover, it explores the future directions for the deep integration of machine learning and hydrogen energy technologies, thereby promoting the low-carbon and intelligent transformation of the shipping industry.

Keywords: machine learning; fuel reforming; hydrogen production; marine energy systems; marine propulsion

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1. Introduction

The global shipping industry, as a key pillar of international trade, is facing increasing pressure to decarbonize. According to the Fourth Greenhouse Gas Study by the International Maritime Organization (IMO), global shipping accounts for approximately 1.076 billion tons of CO_2 emissions annually, representing 2.89% of the total global anthropogenic carbon emissions, and this share continues to rise. Without control measures, shipping emissions could reach 130% of the 2008 levels by 2050. The IMO has set a target to achieve net-zero greenhouse gas emissions in the maritime sector by 2050. To achieve this goal, the IMO has implemented regulations such as the Energy Efficiency Existing Ship Index (EEXI) and the Carbon Intensity Indicator (CII), which aim to drive the shipping industry towards low-carbon and zero-carbon operations [1].

In this context, finding clean and sustainable alternative fuels has become a core issue for the sustainable development of the shipping industry. Hydrogen energy, which produces only water as a byproduct in combustion or electrochemical reactions, is widely regarded as one of the most promising energy carriers for achieving deep decarbonization

in shipping, especially if produced from renewable energy sources. This enables the potential for "zero-carbon emissions" throughout the entire lifecycle from production to propulsion [2,3]. Hydrogen energy, produced via different pathways, exhibits significant differences from traditional marine fuels in terms of lifecycle greenhouse gas emissions and Global Warming Potential (GWP). Wind power-driven electrolysis of hydrogen results in greenhouse gas emissions, which is one of the cleanest hydrogen production pathways [4]; Compared to fossil fuel baselines, the GWP of green hydrogen can be reduced by over 75% [5]; Some studies predict that by 2050, hydrogen produced via electrolysis from grid power could further reduce emissions to a level where just a few grams of CO_2 are emitted for every ton of cargo transported for one nautical mile [6]; In contrast, hydrogen produced via steam methane reforming (SMR) without carbon capture and storage (CCS) results in CO_2 emissions of approximately 10–12 kg per kilogram [4]; Blue hydrogen, before carbon capture, emits 9 to 11 times the amount of CO_2 for each hydrogen unit produced, although Carbon Capture and Storage systems can achieve a carbon capture rate of 85–95% [5]; As for traditional marine fuels, ultra-low sulfur fuel oil (ULSFO) generates approximately 3.613 kg CO_2 eq per kilogram of fuel over its entire lifecycle, which is about 3.7 times that of green hydrogen and 6 times that of green ammonia [7], while heavy fuel oil (HFO), marine diesel oil (MDO), and other fossil fuels contribute to global warming, with a lifecycle greenhouse gas emission intensity of approximately 0.665 kg CO_2 eq per kilowatt-hour of engine output [5], Emission evaluation values for traditional fuels like HFO, marine gas oil, and liquefied natural gas (LNG) vary by study [8].

The application of hydrogen energy in ship propulsion systems mainly involves two technological pathways: hydrogen fuel cells and hydrogen internal combustion engines. Proton exchange membrane fuel cells (PEMFC) and solid oxide fuel cells (SOFC), in particular, efficiently convert hydrogen's chemical energy into electrical energy, offering advantages such as high efficiency, low noise, and zero emissions, making them highly suitable for auxiliary power or propulsion systems on ships [9,10]. Research has shown that PEMFC systems exhibit excellent energy efficiency and power output for ships, demonstrating significant potential for application [11]. This potential has been quantified and validated through simulation models based on real ship operation data. Studies show that the total efficiency of PEM fuel cell-battery hybrid systems can reach 54%, significantly higher than that of hydrogen internal combustion engine-battery hybrid systems (49%) [12]. Another study on a 25,000 t chemical tanker achieved an energy efficiency of 60.56% after integrating and optimizing systems like SOFC [13]. Hydrogen fuel cell systems are now capable of meeting power demands for vessels ranging from small to large. For example, China's hydrogen-powered vessel "Three Gorges Hydrogen Bost No. 1" is equipped with a 500 kW PEMFC system [9]. For larger vessels, design applications are more prominent. One study designed a liquid hydrogen hybrid propulsion system for a 2–MW–class tugboat [14], while another proposed a system integrating SOFCs with a net output power of up to 4743.81 kW [13]. Simulation studies show that a hybrid system integrating a 300 kW fuel cell and lithium batteries can reduce daily CO_2 emissions by 4886 kg compared to traditional diesel propulsion [11]. Additionally, the propulsion system design for the large liquefied hydrogen transport vessel "JAMILA" also verified the feasibility of hydrogen fuel technology [15]. These data confirm the direct environmental benefits of hydrogen fuel cell technology in achieving "near-zero carbon emissions" during the operational phase of ships.

On the other hand, hydrogen internal combustion engines, which are retrofitted from traditional internal combustion engines to directly burn hydrogen or hydrogen-based blended fuels, are also considered a feasible transitional solution, particularly for large oceangoing vessels. For instance, introducing hydrogen into ammonia-diesel dual-

fuel marine engines can effectively optimize the combustion environment, boost thermal efficiency, and sharply cut unburned ammonia and N_2O emissions, with a minor rise in NO emissions that is manageable via after-treatment [16]. Quantitative studies indicate that optimized hydrogen hybrid power systems can achieve significant emission reductions. For example, a hybrid propulsion system for a certain vessel can reduce hydrogen consumption by approximately 10% compared to traditional methods [17], while hydrogen fuel cell systems can achieve zero CO_2 emissions during operation [18]. A case study on container ships shows that an ammonia-fueled hybrid power system can reduce CO_2 emissions by over 2 million kilograms, achieving a reduction rate of 79.36% to 79.51% compared to traditional systems [19]. A life cycle assessment of inland vessels shows that the emission reduction potential of hydrogen-powered technology fluctuates drastically between-6.3% and 85.7%, depending on the hydrogen production pathway [20]. These specific figures demonstrate that emission reduction effects can be measured not only as percentages but also precisely in terms of fuel consumption in kilograms or carbon emissions in tons, providing a more refined scale for evaluating the environmental benefits of different technological paths.

However, the application of hydrogen energy in maritime transport still faces multiple challenges, with technological bottlenecks in every stage of hydrogen production, storage, transportation, and utilization. In the "hydrogen production" stage, although the ultimate goal is to produce "green hydrogen" through water electrolysis driven by renewable energy such as wind and solar power, the cost and economic feasibility remain the major obstacles to large-scale application [21]. By contrast, hydrogen production via fuel reforming based on fossil fuels or biomass—particularly the generation of so-called "blue hydrogen" through integration with carbon capture, utilization, and storage (CCUS) technologies—is currently regarded as a more cost-effective and practically feasible transitional pathway [22]. Data shows that in 2020, China's hydrogen production was approximately 34.1 million tons, with 96% coming from fossil fuels (coal, natural gas, and oil), while water electrolysis accounted for only 4% of the total production [4]. Shipboard fuel reforming hydrogen technologies, such as steam methane reforming, water—gas shift reactions (WGS), and the reforming of liquid fuels like methanol and ethanol, can leverage existing fuel infrastructure to mitigate challenges such as the high cost of green hydrogen production and the lack of storage and transport infrastructure to some extent [23], as shown in Figure 1.

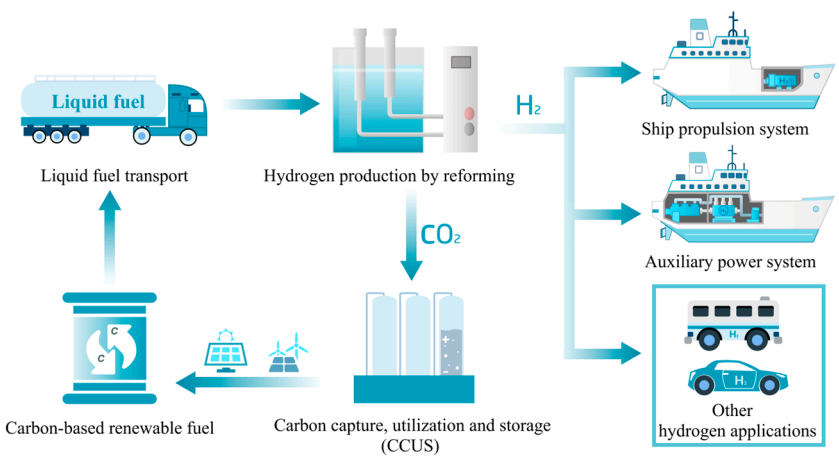

Figure 1. Schematic of hydrogen production process via liquid fuel reforming.

The main chemical equations for fuel reforming hydrogen production include Steam Methane Reforming:

$$CH_4 + H_2O \rightarrow CO + 3H_2 \Delta H = +206\ kJ/mol \quad (1)$$

Water–Gas Shift Reaction:

$$CO + H_2O \rightarrow CO_2 + H_2 \Delta H = -41 \text{ kJ/mol} \quad (2)$$

Methanol Steam Reforming:

$$CH_3OH + H_2O \rightarrow CO_2 + 3H_2 \Delta H = +49 \text{ kJ/mol} \quad (3)$$

Ethanol Steam Reforming:

$$C_2H_5OH + 3H_2O \rightarrow 2CO_2 + 6H_2 \Delta H = +173.6 \text{ kJ/mol} \quad (4)$$

These reactions are essential pathways for hydrogen production, and their efficiency and selectivity are highly dependent on the catalyst and process conditions, which are the key areas where ML can play an optimizing role [24,25]. Despite the advantages of fuel reforming hydrogen production technologies, the processes themselves are highly complex, involving strong non-linear characteristics, multi-variable coupling effects, and thermodynamic and kinetic equilibrium constraints. Traditional trial-and-error methods and first-principle-based mechanistic models are often time-consuming, costly, and difficult to fully capture the intricate underlying patterns in the process when optimizing catalyst formulations, reactor designs, and process parameters. In recent years, the rapid development of Artificial Intelligence (AI) technologies, especially ML methods, has provided new tools for addressing the challenges of modeling and optimizing complex systems. ML models can automatically learn hidden patterns and correlations from large amounts of experimental or simulated data, thereby building high-precision predictive models and guiding optimization decisions [26].

The core contribution of this review is the first systematic, deep cross-analysis and correlation of ML methods, fuel reforming hydrogen production processes, and the unique engineering constraints of marine propulsion systems. The focus is on how ML addresses key challenges in shipboard scenarios: (1) Addressing spatial constraints: Optimizing catalyst activity and selectivity, as well as reactor design using ML, to maximize hydrogen production rates and system energy efficiency within limited space [27,28]. (2) Meeting safety and dynamic load requirements: Using ML to establish high-precision predictive models and real-time intelligent control strategies to ensure stable and safe operation of the reforming system under varying ship operating conditions and complex marine environments, and to facilitate rapid response to changes in power demand [29]. (3) Compliance with IMO regulations (EEXI/CII): Conducting multi-objective collaborative optimization of the entire system using ML to meet emission limits while pursuing minimal fuel consumption and lifecycle costs, directly supporting the ship's compliance with EEXI and CII requirements [30,31].

In the field of fuel reforming hydrogen production, the application of ML technologies is showing tremendous potential. Its advantages, closely aligned with maritime applications, are reflected in several key areas: (1) Accelerating Catalyst Design and Screening: By establishing quantitative relationships between catalyst composition, structural features, and catalytic performance, ML can leverage algorithms such as Random Forest and Gradient Boosting Decision Trees (GBDT) to conduct high-throughput data mining and model prediction, precisely identifying the optimal catalyst formulation. This is crucial for developing efficient and stable catalysts suitable for marine environments. (2) Constructing Accurate Reaction Process Models: ML models are capable of fitting complex reaction kinetics and predicting product distribution and reaction rates under various operational conditions. Their computational speed is typically much faster than traditional mechanistic

models [32]. This is highly significant for achieving rapid dynamic response and real-time optimization control in onboard reforming hydrogen production systems. (3) Optimizing Reactor and System Processes: By integrating computational fluid dynamics (CFD) simulations with process simulation data, ML can optimize reactor design, operational parameters, and the integration and control of the entire hydrogen production system. This maximizes energy efficiency while minimizing costs [28]. Onboard hydrogen production units must adapt to the constraints of limited space, safety, and operational economy aboard vessels, which is critical for meeting regulations such as EEXI and CII [33]. However, the application of hydrogen energy in ship propulsion systems also faces technical and economic challenges: its low volumetric energy density leads to large hydrogen storage systems that challenge ship space capacity [14]; the cryogenic storage requirements for liquid hydrogen are high, and compressed hydrogen presents safety risks; hydrogen refueling infrastructure is severely lacking, and current green hydrogen production costs are prohibitively high [34]. Therefore, developing efficient, compact, and reliable onboard or near-port fuel reforming hydrogen production technologies is of significant practical importance, serving as a bridge between existing fossil fuel infrastructure and future pure hydrogen energy systems.

To ensure the comprehensiveness and representativeness of this review, we employed a systematic literature selection process. First, we conducted searches in academic databases such as Web of Science and Engineering Village, using keywords like "machine learning," "artificial intelligence," "hydrogen production," "fuel reforming," "steam reforming," "catalyst design," and "process optimization," along with their combinations, for literature published between 2015 and 2025. The initial search results were filtered by title and abstract, focusing on original research and review articles that applied ML to the optimization of hydrogen production catalysts, reaction processes, or systems. Building on this, we further refined the selection based on the relevance of the research themes to ship applications, the innovation of the ML methods, and the journal's impact factor. Ultimately, we chose the most timely and authoritative core literature as the primary basis for this discussion. These selected studies cover the full spectrum, from the microscopic design of catalysts to the optimization of ship system integration. The research widely employed Python 3.13.5 and its scientific computing libraries [28,35].

This review systematically conducts a deep cross-analysis and correlation of ML methods, fuel reforming hydrogen production processes, and the unique engineering constraints of marine propulsion systems. The aim is to comprehensively summarize and assess the latest advancements in the application of ML technologies in the fuel reforming hydrogen production field, with a particular focus on its potential in clean energy solutions for the shipping industry. The paper will explore three key areas; (1) ML-Driven Catalyst Research and Design Optimization: Emphasizing its application in performance prediction, high-throughput screening, and synthesis optimization; (2) ML-Assisted Reaction Process Modeling: Covering the development of kinetic models, multi-scale simulation integration, and the analysis of reaction mechanisms for interpretability; (3) ML in Equipment Design and System Process Optimization: Including reactor structure optimization, multi-objective collaborative optimization of process parameters, performance prediction and monitoring, as well as intelligent control strategies tailored for marine propulsion systems. By reviewing these cutting-edge studies, this paper aims to provide researchers and engineers in the maritime field with a clear technological roadmap, driving the development of efficient, intelligent onboard hydrogen production systems empowered by ML, and accelerating the decarbonization and intelligent transformation of the shipping industry.

2. Catalyst Design and Optimization via Machine Learning

Catalysts are the core of fuel reforming hydrogen production technologies, and their performance directly impacts reaction conversion rates, product selectivity, energy efficiency, and the long-term stability of the system. Traditional catalyst development relies on trial-and-error methods and the accumulated experience of researchers, which is both time-consuming and costly. This traditional paradigm is inefficient: for a complex multi-component catalyst system, the cycle from preliminary design to laboratory validation is lengthy. While first-principle-based computational screening has a theoretical foundation, calculating key parameters like adsorption energy for a single catalyst model can take hundreds of hours. In contrast, once an ML model is trained, performance prediction for thousands of candidate catalyst compositions can be completed in seconds or minutes [36]. The introduction of ML technologies provides a powerful tool for accurately predicting catalyst performance and achieving high-throughput screening, significantly accelerating the research and development process. This section focuses on the latest advancements in ML applications for catalyst research and design optimization in fuel reforming for hydrogen production, covering three main areas: catalyst performance prediction, high-throughput screening and the design of new catalysts, and optimization of synthesis conditions and active site regulation. Table 1 presents typical ML application studies in the design of fuel reforming hydrogen catalysts, covering various hydrogen production methods, ML models, input-output variables, and model performance indicators. Detailed data can be found in the Appendix A.1.

Table 1. Research on Catalysts and Design Optimization Driven by Machine Learning for Hydrogen Production via Fuel Reforming.

Hydrogen Production Methods	Machine Learning Approaches	Inputs	Outputs	Model Performance	References
Methane Dry Reforming	Interpretable Machine Learning (IML), utilizing SHapley Additive exPlanations (SHAP) and Partial Dependence Values (PDV) tools	Catalyst Parameters (Type of Promoter, Support Properties), Reaction Conditions (Temperature, Fuel-to-Oxidant Ratio, Space Velocity)	Methane Conversion Rate, Carbon Dioxide Conversion Rate, Hydrogen Production Rate, Carbon Deposition Amount	The model exhibits high prediction accuracy, with experimental validation showing an R^2 value greater than 0.9 and a low RMSE	[36]
Methane Dry Reforming	An explainable CatBoost model, enhanced with interpretability tools, to improve transparency	Reaction temperature, gas hourly space velocity (GHSV), calcination temperature, nickel loading, and catalyst structural parameters	Methane Conversion Rate	The CatBoost model predicts the methane conversion rate with an R^2 value of 0.91, and experimental validation shows an error of less than 10%	[37]
Catalytic Steam Reforming of Biomass Tar	Machine Learning Algorithms, Random Forest (RF), Artificial Neural Networks (ANNs) for Optimizing Nickel-Based Catalysts	Reaction Temperature, Catalyst Support, Additive Type, Nickel Loading, Calcination Temperature	Toluene Conversion Rate	The RF model for predicting toluene conversion rate was evaluated with an accuracy of 0.99 and an AUC of 0.92	[38]
Carbon Dioxide Methanation	Explainable Machine Learning (SHAP Analysis), XGBoost	Reaction temperature, nickel content, calcination temperature, particle size, reduction time	Carbon dioxide conversion rate, methane selectivity	The XGBoost model demonstrates high prediction accuracy, as verified through experimental validation	[39]

Table 1. *Cont.*

Hydrogen Production Methods	Machine Learning Approaches	Inputs	Outputs	Model Performance	References
Dry Reforming of Methane (CO_2 Methanation)	Multilayer Perceptron (MLP) and Nonlinear Autoregressive Exogenous (NARX) Neural Networks	Calcination temperature, reduction temperature, reaction temperature, reaction time, and Ni loading	Methane conversion rate, carbon dioxide conversion rate	The NARX neural network achieves the highest R^2 = 0.998 and the lowest MSE = 3.24×10^{-9}	[40]
Sorption-Enhanced Chemical Looping Reforming	Artificial Neural Network	Operating Parameters (Temperature, Pressure, Flow Rate) and Catalyst/Adsorbent Properties	Methane Conversion Rate, Hydrogen Purity, and Carbon Dioxide Removal Efficiency	The ANN model demonstrates high prediction accuracy, $R^2 \geq 0.9889$	[41]

2.1. Prediction of Catalyst Performance

Catalyst performance prediction using ML models involves using input features such as the catalyst composition, structure, synthesis parameters, and reaction conditions to predict key performance indicators like activity, selectivity, and stability. As one of the most fundamental and widely applied research scenarios for ML in the catalyst field, the core focus is on constructing accurate structure-performance correlation models for catalysts. The ultimate goal is to support the rational design of catalysts, reducing reliance on traditional trial-and-error methods.

2.1.1. Key Inputs and Outputs of the Prediction Model

The first step in constructing a high-performance prediction model is to define its inputs and outputs. The input features primarily include the following categories:

- Catalyst characteristics, such as the type of active metal, types and contents of promoters/activators, specific surface area, pore volume, pore size distribution, and crystallite size, etc. [36,42].
- Catalyst synthesis parameters, such as preparation method, precursor type, calcination temperature, reduction temperature, and stirring time, etc.
- Reaction operating conditions, such as reaction temperature, pressure, feed flow rate, feed gas composition, and space velocity, etc. [43,44].

The output targets of the model, or the performance indicators to be predicted, usually include:

- Activity indicators, such as methane conversion rate (X_CH_4), carbon dioxide conversion rate (X_CO_2), and carbon monoxide conversion rate (X_CO), as well as the space-time yield (STY) for processes like WGS and methanol synthesis, etc.
- Stability indicators, such as catalyst deactivation rate and the retention of activity after a specific period, etc. [44].
- Other performance indicators, such as hydrogen production rate and syngas (H_2/CO) ratio, etc.

2.1.2. Mainstream Machine Learning Algorithms and Their Applications

Many studies have applied various machine learning algorithms to construct prediction models, with ANN, RF, and gradient boosting decision trees being the most commonly used.

Artificial Neural Networks, with their strong nonlinear modeling capability, are well-suited for complex catalytic reactions and have been widely applied in various reforming processes. For instance, Hossain et al. used ANNs to predict the performance of nickel-based catalysts in the direct methane reforming reaction [44]. Ayodele et al. analyzed the

parameter correlations of Co/Pr_2O_3 catalysts for Dry Reforming of Methane (DRM) [45]. Byun et al. developed a model that integrates methanol reforming technology with environmental and economic indicators [46], while Song et al. used ANNs to reverse-engineer the optimal nickel/aluminum loading and preparation process for ethanol steam reforming catalysts [47]. In fuel reforming hydrogen production research using ANNs, common input parameters include temperature, pressure, steam-to-carbon ratio, steam temperature, pipe diameter, gas flow rate, heat transfer coefficient, specific surface area, and support material type. Output parameters typically include methane conversion rate, carbon monoxide conversion rate, hydrogen selectivity, carbon monoxide selectivity, and space-time yield, as shown in Figure 2.

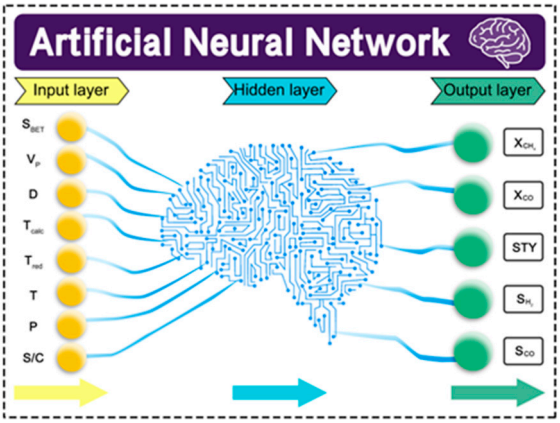

Figure 2. Artificial Neural Networks in the Fuel Reforming Hydrogen Production Process.

Ensemble learning methods, such as RF and XGBoost, have gained popularity in recent years due to their high accuracy, fast training speed, and ability to handle high-dimensional data, as shown in Figure 3. XGBoost is an optimized gradient-boosted decision tree framework, featuring regularized overfitting suppression, efficient parallelization, and high predictive performance. A typical example of this approach is the study conducted by Pérez-Ramírez et al. They compiled a literature dataset containing approximately 1425 data points and applied advanced machine learning algorithms, such as XGBoost, random forests, and gradient boosting decision trees (GBT), to predict the space-time yield of methanol synthesis from CO_2 hydrogenation. However, their study also revealed a common challenge in this field: even well-performing algorithms may not perform adequately when validated with new experimental data points. This indicates that the model requires additional training on new data and that its generalization capability needs further improvement [48].

In addition to prediction accuracy, the interpretability of the model is crucial for understanding catalytic mechanisms. Tools such as SHAP and PDP are used to explain model decisions and identify key descriptors. For example, the IML framework developed by Roh et al. utilizes tools like SHAP to predict the performance of DRM catalysts and quantifies the impact of various input features on catalyst performance, providing guidance for experimental validation [36]. Similarly, in the optimization of $Cu/ZnO/Al_2O_3$ catalyst synthesis, SHAP analysis revealed the key influence of synthesis parameters on copper surface area [49]. As shown in Figure 3, explainable AI, through the iterative cycle of "data—model—experiment—knowledge", is driving catalyst development toward a more intelligent and scientific direction. ANN and ensemble learning are core algorithms for performance prediction, and the integration of explainable tools achieves the dual goals of "high-precision prediction + mechanistic analysis," providing support for the rational design of catalysts.

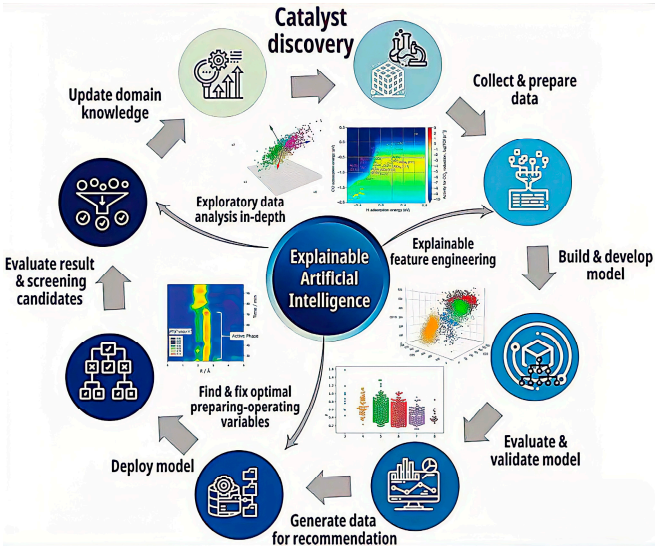

Figure 3. Application Process of Interpretable Artificial Intelligence in Catalyst Discovery [50].

2.1.3. Predictive Studies for Different Reaction Systems

At present, machine learning prediction models have been widely applied in various mainstream fuel reforming hydrogen production processes, demonstrating their extensive applicability.

In the field of methane DRM research, Alotaibi et al. integrated computational fluid dynamics, artificial neural networks, and multi-objective genetic algorithms (MOGA) to enhance the DRM process. They first generated reliable data through CFD simulations, which were then used to train an ANN model to establish correlations between yield, conversion rate, flow rate, carbon deposition, and input parameters. The trained ANN model was subsequently used as the objective function for MOGA to identify optimized input parameters that maximize conversion rate and yield while minimizing carbon deposition [51]. Du et al. employed high-throughput experiments (HTE) combined with ML to obtain an unbiased dataset and proposed catalyst design guidelines [52].

The water–gas shift reaction is crucial for hydrogen purification. Kim et al. compared the performance of an ANN model for predicting the behavior of Pt-based catalysts in WGS reactions [53]. They further conducted machine learning-based high-throughput screening, strategic design, and knowledge extraction to develop $Pt/Ce_XZr_{1-X}O_2$ catalysts [54]. One study also compared deep learning models applied to WGS catalysts for hydrogen purification [55].

In the field of diesel reforming, Liang et al. developed a regression model correlating reaction conditions and hydrogen yield in diesel reforming reactions using machine learning-assisted Aspen Plus simulations. They found that temperature is a critical factor, while the steam-to-carbon ratio plays a crucial role in enhancing baseline yield [56].

CO_2 hydrogenation to methanol is an important approach that combines CO_2 utilization with hydrogen carrier methanol production. Asif et al. reviewed the application of ML in CO_2 catalytic hydrogenation to methanol, exploring its potential in process optimization, performance prediction, reaction kinetics modeling, and enhancing catalyst activity [57]. Another study applied machine learning algorithms, such as ANN, SVR, and GPR, in combination with principal component analysis (PCA) to predict the performance of copper-based catalysts in this reaction [58].

Performance prediction models constructed using algorithms such as ANN and Random Forest have enabled precise forecasting of catalyst activity, selectivity, and stability. These models provide efficient screening tools for different reforming reaction systems, signifi-

cantly reducing trial-and-error costs. Catalyst performance prediction offers a solid basis for identifying screening targets, while the integration of high-throughput screening with ML further addresses the core challenge of "efficient screening," ultimately reshaping the catalyst development paradigm. This combination accelerates the discovery and optimization of high-performance catalysts, leading to more cost-effective and faster development cycles.

2.2. High-Throughput Screening and the Design of Novel Catalysts

The integration of high-throughput screening (HTS) and machine learning is reshaping the paradigm of catalyst development. The traditional "trial-and-error" approach, which is time-consuming, labor-intensive, and heavily reliant on experience, is gradually being replaced by data-driven design [59]. ML has significantly accelerated the process of identifying high-performance materials from candidate catalyst libraries and uncovering their underlying design principles by mining hidden "structure–activity" relationships from vast amounts of experimental or computational data [60].

2.2.1. Synergistic Screening Based on First-Principles Calculations and Machine Learning

Density Functional Theory (DFT) calculations can provide accurate descriptors of catalyst electronic structures, such as adsorption energy and d-band center, but their high computational cost makes them difficult to be directly used for large-scale screening. Machine learning models, trained on small sample datasets from DFT calculations, can rapidly predict the catalytic performance of other candidate systems. In the context of CO_2 reduction reactions, researchers combined DFT and ML techniques for high-throughput screening of Cu-Mn-Ni-Zn high-entropy alloy catalyst systems. They found that when Mn atoms served as the active sites, the catalyst exhibited excellent activity and selectivity for methane and methanol production [61]. Pandit et al. successfully designed bimetallic and trimetallic catalysts for the hydrogen evolution reaction (HER) using supervised machine learning, demonstrating the powerful potential of this synergistic strategy in catalyst design [62].

2.2.2. Closed-Loop Optimization of High-Throughput Experimentation and ML

High-throughput experimentation platforms enable the parallel synthesis and testing of a large number of catalyst samples, generating standardized and scalable datasets that provide high-quality training data for machine learning models. Du et al. combined high-throughput experimentation with machine learning to obtain an unbiased dataset for methane dry reforming and derived a catalyst design methodology from the data. This study successfully moved away from the traditional trial-and-error approach, which relies on prior assumptions, and achieved data-driven catalyst development [52]. In the field of biomass catalytic pyrolysis for hydrogen production, Persaud et al. employed machine learning to guide the optimization of nickel-based catalysts, significantly enhancing the yield of biohydrogen. This demonstrated the effectiveness of machine learning in optimizing catalysts for complex reaction systems [27].

2.2.3. Design of High-Entropy Alloys and Multicomponent Catalysts

High-entropy alloy catalysts, with their vast compositional space and tunable electronic structures, have become a frontier in the field of catalysis. However, their complexity presents challenges that traditional methods struggle to address. ML has emerged as a key tool for exploring such extensive material spaces. Rittiruam et al. utilized first-principles DFT and machine learning techniques to conduct high-throughput screening of high-entropy alloy catalysts for CO_2 reduction reactions. They identified the optimal catalyst compositions for CH_4 and CH_3OH selectivity and revealed that Mn serves as the active site, with Cu/Ni/Zn as neighboring atoms. These findings provide new insights

into understanding the CO_2 reduction behavior of such complex alloy catalysts [61]. These studies demonstrate that ML can effectively handle high-dimensional features, enabling the identification of promising multicomponent catalyst formulations from a vast range of possibilities.

2.2.4. Identification and Design of Catalyst Active Sites

Identifying the active sites of a catalyst is central to designing efficient catalysts. ML aids researchers in recognizing key active sites and their structural features by analyzing the complex relationships between structural characterization data and performance data. The special quasi-random structure (SQS) method, proposed by Zunger et al., provides the theoretical foundation for generating representative alloy models, significantly advancing the application of ML in alloy catalyst design [63].

DFT + ML collaborative screening, high-throughput experimentation combined with ML-based closed-loop optimization, and other methods have overcome the limitations of traditional trial-and-error approaches, accelerating the development and active site identification of High Entropy Alloys (HEAs) and multi-component catalysts. Once high-performance catalyst candidates are identified through high-throughput screening, the precise optimization of synthesis conditions and the regulation of active sites become crucial for enhancing the catalyst's practical performance in real-world applications. These advanced methodologies streamline the catalyst development process, ensuring that the most promising candidates are efficiently optimized for industrial use.

2.3. Optimization of Synthesis Conditions and Active Site Regulation

The final performance of a catalyst depends not only on its chemical composition but also on its microstructure, including specific surface area, pore structure, crystallite size, and metal dispersion. This microstructure is closely related to the synthesis conditions, such as the precursor, calcination temperature, and stirring time. Machine learning enables precise control and optimization of the catalyst synthesis process by establishing complex nonlinear mappings between "synthesis parameters, structural features, and catalytic performance".

2.3.1. Global Optimization of Synthesis Parameters

Artificial neural networks, with their powerful ability to model nonlinear relationships, are widely used to optimize multiple parameters in the catalyst synthesis process. Figure 4 illustrates the full automation of the process, from data preparation to optimal variable identification, using deep neural networks [28]. In their study of Ni/Al_2O_3 catalysts for ethanol steam reforming, Song et al. constructed several neural network models. Among them, the NN-2 model used preparation methods (co-precipitation, precipitation, and impregnation) as well as the loading amounts of Ni and Al as inputs, with ethanol conversion rate, H_2 yield, and selectivity as outputs. This model successfully predicted that co-precipitation with Al loading of 42.49% and Ni loading of 12.35% provided the optimal conditions, achieving the conversion rate of ethanol and the selectivity of hydrogen reached 79.6% and 91.4%, respectively [47]. Cursaru et al. used artificial neural networks to analyze the relationship between the synthesis conditions of Co/MCM-48 catalysts and their performance in steam reforming reactions, demonstrating the value of machine learning in optimizing synthesis formulations [64].

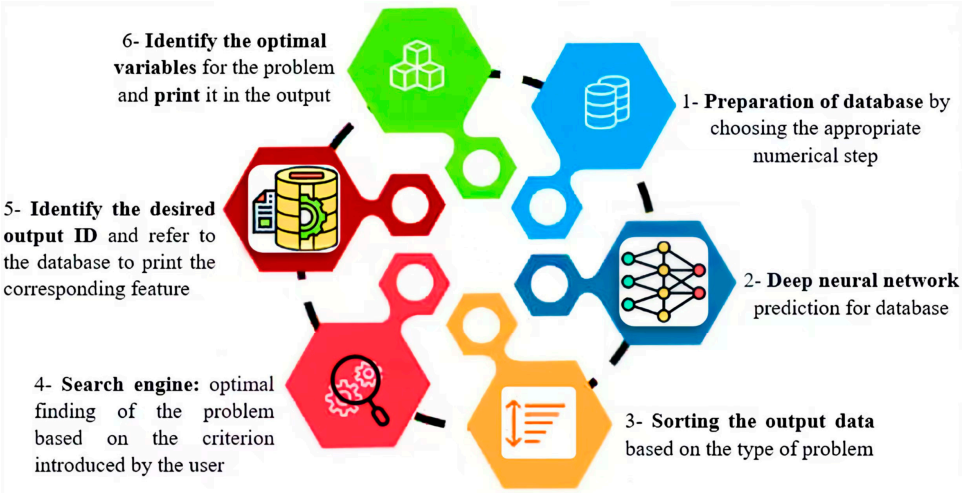

Figure 4. Flowchart of the Optimization Problem-Solving Process Based on Neural Networks and Database Integration [28].

2.3.2. Prediction and Control of Microstructure

Microstructural parameters of a catalyst, such as specific surface area, pore size distribution, and crystallite size, are key determinants of its performance. ML models can predict these structural characteristics of catalysts under specific synthesis conditions. Song et al.'s NN-1 neural network model used preparation methods and metal loading amounts as inputs and successfully predicted the specific surface area, pore volume, pore size, and crystallite size of the final catalyst [47]. This predictive capability allows researchers to perform "virtual screening" of the catalyst's physical structure before actual synthesis, significantly reducing the experimental workload.

2.3.3. Interpretable Machine Learning for Revealing Key Regulatory Factors

In synthesis optimization, IML tools can reveal key controlling factors. When optimizing $Cu/ZnO/Al_2O_3$ catalysts, Random Forest combined with SHAP analysis showed that the Cu/Zn ratio was the most critical parameter [28]. SHAP, as a well-recognized interpretability analysis framework for black-box machine learning models, is capable of quantifying the importance of input features, explicitly revealing the intrinsic variable regulation mechanisms of the model, and effectively enhancing the transparency of the model's decision-making process. Roh et al. also used SHAP/PDV tools to analyze the performance regulation mechanisms of DRM catalysts, which not only improved the model's reliability but also deepened the understanding of the catalyst performance regulation mechanisms [36]. IML has revealed the regulatory patterns of synthesis parameters, providing clear experimental guidance for the directed synthesis of marine hydrogen production catalysts.

2.3.4. Regulation of the Electronic Structure of Active Sites

On the basis of understanding the macro key factors, the micro-regulation of the electronic structure of active sites represents a higher-level design. ML can design active sites with ideal electronic states by correlating synthesis conditions, structural features, and final electronic properties. In methanol steam reforming, the Pt/In_2O_3 catalyst exhibits excellent performance [65]. In addition, machine learning-assisted analysis of metal-support interactions and other phenomena has become a research hotspot [65]. In the future, the combination of DFT-calculated descriptors and machine learning models will enable precise design and regulation of the electronic structure of active sites, leading to enhanced activity and improved resistance to carbon deposition.

2.3.5. Inverse Design Framework

ML not only enables forward prediction of synthesis outcomes but also offers the more powerful capability of "inverse design". Given target performance, such as high H_2 yield and high stability, ML can recommend the optimal synthesis formulations and conditions. Although this is a complex challenge, it marks a shift in catalyst development from being "experience-driven" to "goal-driven" [36].

Machine Learning, through global optimization of synthesis parameters, microscopic structure prediction, and the use of explainable tools, has constructed a comprehensive "synthesis-structure-performance" control system, supporting catalyst reverse design and directed synthesis. By integrating ML models with experimental and simulation data, a more accurate understanding of complex reaction dynamics can be achieved, enabling the optimization of reaction conditions and the development of highly efficient hydrogen production systems. This approach helps overcome the challenges of scaling up and improving catalytic performance under real-world conditions. High-performance catalysts provide the material foundation for hydrogen production, but the multi-scale coupling issues in the reaction process require precise model support.

3. Reaction Process Modeling Aided by Machine Learning

3.1. Chemical Reaction Kinetics Models

Chemical reaction kinetics models are essential for understanding and optimizing fuel reforming processes, as their accuracy directly impacts reactor design, process optimization, and system control. Traditional kinetic models are typically based on empirical or semi-empirical equations, relying on simplified assumptions and experimental fitting. These models face limitations when dealing with complex reaction networks, multicomponent systems, and nonlinear behaviors. In recent years, the introduction of machine learning techniques has provided new methods for kinetic modeling, enhancing the predictive capability, generalization, and computational efficiency of models through data-driven approaches [65].

3.1.1. ML Replacement and Enhancement of Traditional Mechanistic Models

Traditional mechanistic models are built based on physical and chemical laws, and while they offer good interpretability, they often sacrifice accuracy due to simplifying assumptions and incur high computational costs. Machine learning models can serve as a substitute or enhancement, significantly improving computational efficiency while maintaining predictive accuracy.

This efficiency improvement is revolutionary. For a steam methane reforming reactor that involves complex multiphase flow and reactions, full-scale computational fluid dynamics simulations can take several days to complete. However, a trained (ML surrogate model, while maintaining a high degree of accuracy in matching CFD simulations ($R^2 > 0.96$), can predict the same conditions in just a few milliseconds [66,67]. For instance, an optimized random forest model in one study was able to provide predictions in approximately 50 milliseconds per instance [67]. This makes real-time optimization, uncertainty analysis, and sensitivity studies of process parameters based on extensive scenario simulations feasible, offering critical technological support for rapid decision-making in response to dynamic loads in onboard systems.

For example, the ANN model developed by Qiu et al. predicts the hydrogen concentration within an SMR reactor using spatial coordinates (r, z) as input. The model shows a high degree of agreement with CFD simulation results, achieving an R^2 value of approximately 0.998 [68]. Aklilu et al. used Gaussian process regression (GPR) to replace traditional microkinetic models, successfully predicting the reaction rate of carbon dioxide

hydrogenation to methanol. This method significantly reduces the computational burden associated with solving complex differential equations [69].

Another strategy is to combine ML with traditional mechanistic models, creating a hybrid modeling framework. In Aspen Plus process simulations, traditional thermodynamic models are unable to fully capture the interactions between parameters. In their optimization of diesel auto-thermal reforming, Huang et al. integrated 12 machine learning models with Aspen Plus and used the XGBoost method for screening and analysis. This approach enhanced the model's ability to describe the interactions in reaction kinetics and significantly improved the accuracy of hydrogen yield predictions [70]. Roh et al. proposed an interpretable ML framework for dry methane reforming, which combines SHAP and PDV tools. This approach ensures high predictive accuracy (R^2 = 0.96) while maintaining model interpretability [36]. The ML surrogate model reduces computational costs, and the hybrid model strikes a balance between accuracy and interpretability, providing a flexible and efficient solution for fuel reforming kinetics modeling.

3.1.2. Kinetic Analysis of Complex Reaction Networks

Fuel reforming processes often involve complex networks of parallel, series, and competitive reactions, which are difficult for traditional kinetic modeling to fully capture. ML techniques, by handling high-dimensional data and nonlinear relationships, have significantly enhanced the analytical capabilities of complex reaction networks.

In multiple reforming reactions, dry reforming, steam reforming, and auto-thermal reforming coexisting systems, ML models have successfully decoded the dominant pathways and rate-determining steps of the reaction network. In the triple reforming (TRM) of biogas for syngas production, researchers employed interpretable ML tools to analyze the factors influencing methane conversion rate, CO_2 conversion rate, and H_2/CO ratio. They extracted heuristic rules for achieving high target values [71]. Similarly, in methanol auto-thermal reforming, ML models analyzed the relationship between operating parameters and product distribution, revealing the competitive dynamics between oxidation and reforming pathways [72].

For dynamic systems where the reaction network evolves over time, ML methods exhibit unique advantages. In packed bed reactors, ML models, combined with CFD data, predicted the spatiotemporal distribution of hydrogen concentration, revealing the coupling effects between local kinetics and mass transfer [52].

The ML surrogate model reduces computational costs, while the hybrid model balances accuracy and interpretability, effectively addressing the limitations of traditional kinetic models in complex reaction networks. While traditional kinetic modeling focuses on reaction rates and network analysis at a single scale, higher-level optimization requires the integration of cross-scale information. Machine learning provides an efficient tool for this purpose.

3.2. Multiscale Simulation and Data Fusion Strategies

ML has significantly enhanced the predictive accuracy and optimization efficiency of fuel reforming hydrogen production processes by integrating multi-level models, from atomic to reactor scales. Multiscale simulations encompass quantum chemical calculations, microkinetics, CFD, and system-level modeling, while data fusion combines experimental, simulation, and high-throughput data to address issues of data sparsity and generalization. These ML methods have accelerated catalyst design, reaction optimization, and process scale-up efforts.

3.2.1. Integration of Atomic-Scale Models and Machine Learning

At the atomic scale, DFT calculations provide the energy and kinetic parameters for catalyst surface reactions, though they come with high computational costs. Machine learning, by training predictive models based on DFT data, enables rapid catalyst screening. Ugwu et al. reviewed the application of DFT and machine learning in heterogeneous catalytic reactions, including methane reforming and steam reforming processes. They highlighted how descriptors such as adsorption energy and d-band center are generated to predict catalyst activity [73]. Kim et al. employed machine learning models based on DFT-calculated transition state energies to screen platinum catalysts for the water–gas shift reaction, thereby reducing the time required for experimental screening [74]. In their study on methanol synthesis from methane and water, Ban et al. used DFT calculations to determine the reaction energy barriers and employed machine learning to screen single-atom vacancy dual active site (SA-FLP) catalysts. They developed a model to correlate catalytic activity with electronic structure descriptors. This work achieved data correlation from the electronic scale to the reactor scale by predicting macroscopic reaction rates through micro-level reaction energy barriers [75]. Additionally, Du et al. combined high-throughput experimentation with machine learning to establish an unbiased dataset for methane dry reforming catalysts. They used models such as decision trees to identify key elemental combinations, including Ni-Li-Al-Nb and Al-Nb-Hf, providing valuable guidance for the design of low-temperature methane dry reforming catalysts [52]. These methods combine atomic-scale simulations with data-driven models, enabling precise prediction of catalyst performance.

3.2.2. Kinetic Modeling at the Reaction Scale and Machine Learning

At the reaction scale, microkinetic models describe the surface reaction network, but parameter calibration is complex. Machine learning accelerates the optimization of kinetic parameters by using surrogate models. Wang et al. proposed an integrated framework combining chemical reaction models, ensemble learning methods, and the whale optimization algorithm for optimizing hydrogen yield in adsorption-enhanced steam methane reforming (SE-SMR). This framework uses an XGBoost surrogate model to replace CFD simulations, enabling hydrogen yield predictions within 4 s. The overall optimization process takes less than 1 min, with an error of only 0.79% [76]. In methane dry reforming, Roh et al. utilized the CatBoost algorithm within an interpretable machine learning framework, combining SHAP and PDV tools to predict the CH_4 conversion rate of catalysts. They recommended high-performance catalysts, Ni-Rh/γ-Al_2O_3 and Rh-Ce/γ-Al_2O_3, and identified key factors influencing catalytic performance through SHAP analysis, including the active metal Ni, reaction temperature, and gas space velocity [36].

3.2.3. Data Fusion Strategies and Cross-Scale Validation

Data fusion involves integrating data from various sources, such as experimental measurements, simulation outputs, and literature data, to address issues of data inconsistency. Sharma and Liu reviewed hybrid physics-informed machine learning methods, where physical laws are embedded into data-driven models to ensure the robustness of these models during scale expansion [77]. Wang et al. developed a surrogate model for hydrogen yield prediction using the XGBoost ensemble learning method in the context of adsorption-enhanced steam methane reforming. Leveraging an operating parameter database generated by a CFD chemical reaction model, this surrogate model achieved high-precision predictions. The key operating parameters included temperature, pressure, velocity, and water–carbon ratio, among others [78]. Data fusion, through the integration of multi-source data, application of physics-informed ML, and data augmentation techniques,

has enhanced the robustness and generalization of multiscale models. The R^2 value of the hybrid ensemble model can reach 0.996 [79].

ML has enabled the integration of cross-scale information from atomic to reaction scales, and by combining data fusion strategies, it has improved the generalization and robustness of the models, providing reliable support for multiscale optimization. While multiscale models address the challenge of "cross-scale information integration", IML further overcomes the "black-box model mechanism" limitation, offering physicochemical support for reaction pathway optimization.

3.3. Interpretable Analysis of Reaction Pathways and Mechanisms

IML provides a new approach for understanding complex catalytic networks [36,80]. The development of IML techniques, combined with density functional theory (DFT) calculations and experimental validation, has opened new pathways for understanding intricate catalytic networks [36,80]. IML tools, such as SHAP and PDV, when integrated with DFT calculations, offer in-depth analysis of reaction networks, active sites, and deactivation mechanisms.

3.3.1. Application of Interpretable Machine Learning Tools

IML tools quantify feature importance to identify key variables that influence reaction pathways. In methane dry reforming, Roh et al. developed an IML framework that combines SHAP and PDV analysis to identify key factors influencing CH_4 conversion rate, including reaction temperature, gas space velocity, and the type of active metal. The model, validated experimentally with 17 different catalysts, accurately predicted the CH_4 conversion rate [36]. For complex reaction networks, Esterhuichen et al. discussed various interpretable machine learning methods, emphasizing the potential of interpretable machine learning in accelerating hypothesis formation and knowledge generation [81]. These tools not only enhance model transparency but also guide experimental design, such as prioritizing catalyst composition selection through SHAP values.

3.3.2. Integration of DFT and Machine Learning for Mechanistic Studies

DFT calculations provide information on reaction energy barriers and intermediates, but they are computationally intensive. Machine learning accelerates mechanistic exploration by training regression models on DFT data. Ugwu et al. applied DFT and machine learning to methane reforming reactions. They used DFT to analyze the catalyst's electronic structure and reaction energy barriers, and combined this with machine learning to screen for high-performance catalysts [73]. This integration approach is particularly well-suited for high-throughput screening. In alloy catalyst screening, Yu et al. used the adsorption energies of C/O formation based on DFT calculations as activity descriptors, combined with clustering algorithms to identify thermodynamically stable alloys, significantly enhancing the screening efficiency [82].

3.3.3. Reaction Pathway Analysis

IML analyzes competitive reaction pathways by visualizing feature contributions. In methanol synthesis, Bhardwaj et al. developed a performance prediction model based on principal component analysis and artificial neural networks. This model provided data support for analyzing key factors influencing reaction pathways and, through sensitivity analysis, identified temperature as the key parameter affecting selectivity [58].

The integration of IML tools with DFT has revealed key variables and mechanisms in reaction pathways, achieving a unified balance between "predictive accuracy" and "mechanistic interpretability". While reaction models uncover underlying patterns, engineering implementation requires addressing equipment and system-related challenges.

4. Machine Learning in Equipment Design and System Process Optimization

Reaction process modeling provides the theoretical foundation for process optimization, while the application of machine learning in equipment design and system-level optimization further translates theory into engineering practice, particularly addressing the compactness and real-time requirements of onboard hydrogen production systems. Beyond its significant potential in catalyst design and reaction modeling, machine learning has also been extended to hydrogen energy system optimization, playing a crucial role in equipment design and process refinement. Traditional physics-based modeling and simulation methods, such as CFD and Aspen Plus, while offering deep insights into the fundamental nature of processes, are often associated with high computational costs and lengthy processing times, making them challenging to apply for rapid optimization and real-time control [59]. Machine learning, by learning complex nonlinear relationships from large-scale process data, constructs high-precision and efficient surrogate models. These models provide powerful tools for reactor design innovation, global optimization of process parameters, system performance prediction, and intelligent control [83]. Table 2 systematically summarizes key studies on ML applications in equipment design and system process optimization for hydrogen production, covering diverse production pathways, ML models, input/output variables, and optimization effects. For more detailed data, please refer to Appendix A.2.

Table 2. Application of Machine Learning in Equipment Design and System Process Optimization.

Application Scenarios	Machine Learning Approaches	Inputs	Outputs	Model Performance	References
Steam Methane Reforming	Deep Neural Networks (DNN) Combined with Random Search Optimization Algorithm	Reactor length, feed flow rate, heat flux, S/C (steam-to-carbon ratio)	Hydrogen yield	The DNN model offers high prediction accuracy and the capability for multi-objective optimization	[28]
Methane Autothermal Reforming	Deep Reinforcement Learning (DRL) combined with Random Forest models and Q-learning algorithm	Oxygen-to-methane ratio, steam-to-methane ratio, temperature	Operating expenses (OPEX), carbon emissions per unit time, hydrogen yield	After DRL optimization, OPEX decreased by 10%, carbon footprint was significantly reduced compared to traditional processes, and hydrogen yield increased by 13%	[79]
Solar-driven Methanol Steam Reforming	Grey Relational Analysis (GRA) and Genetic Algorithm Optimized BP Neural Network (GA-BPNN)	Reaction temperature, methanol flow rate	Methanol conversion rate, hydrogen yield, carbon monoxide selectivity	The GA-BPNN model demonstrates high prediction accuracy, outperforming traditional prediction models	[84]
Fuel Cell and Ammonia-Hydrogen Internal Combustion Engine Hybrid System	Integrated Deep Learning (PCA, MCNN, SVM)	Voltage, current, temperature, pressure, fault signals	Fault diagnosis accuracy, system reliability	Overall diagnostic accuracy of 98.15%, with multi-system fault diagnosis at 96.67%	[85]
Biomass Gasification for Hydrogen Production	Gradient Boosting Regression Model combined with Particle Swarm Optimization (PSO)	Biomass type, temperature, reaction time, biomass concentration, pressure, reactor type	Hydrogen yield	After PSO, the test R^2 of the Gradient Boosting Regression (GBR) model is 0.958, and its cross-validation R^2 is 0.917	[86]

The implementation process of machine learning in improving efficiency, reducing energy consumption, and meeting performance requirements during design and operation consists of five main stages:

Phase 1 involves constructing the dataset, which includes collecting experimental data, simulation data, literature data, and operational data from the ship, providing the foundation for the ML model.

Phase 2 focuses on building the ML model, encompassing key steps such as model architecture design, hyperparameter optimization, and model training to ensure the model's efficiency and accuracy.

Phase 3 involves multi-objective collaborative optimization of design and operational parameters, further enhancing the overall performance of the system.

Phase 4 is the evaluation and comparison of performance. Based on whether the system meets IMO regulations, it will proceed to Phase 5 for deployment and integration, or continue iterative optimization and model updates to ensure the system consistently meets design requirements, as shown in Figure 5.

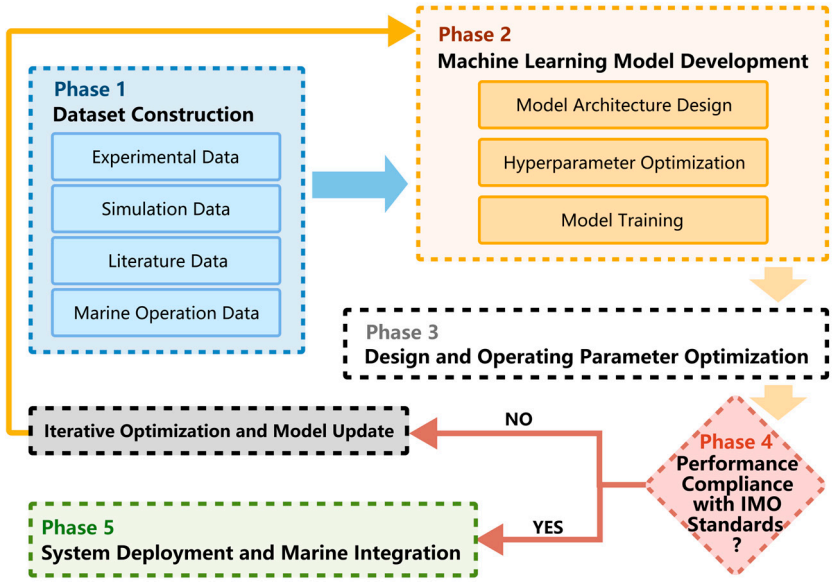

Figure 5. Machine Learning-Based Onboard Reforming Hydrogen Production Equipment and Process Optimization Diagram.

4.1. Reactor Structure Optimization

The fuel reforming hydrogen production reactor, as the core equipment of onboard hydrogen production systems, plays a crucial role in determining the mixing of reactants, heat and mass transfer efficiency, catalyst utilization, and product distribution. ML provides an efficient pathway for structural optimization. By integrating ML with traditional multiphysics simulations, an effective research paradigm for optimizing reactor structure has been developed. For industrial reactors with a hydrogen production capacity of 100–1000 Nm^3/h, the typical objective functions and constraints for methanol steam reforming optimization design are as follows:

The comprehensive optimization objective function is defined to maximize the weighted sum of energy conversion efficiency and hydrogen production rate per unit volume, expressed as

$$Max\ J = W_1\eta + W_2 Y_{H_2} \tag{5}$$

where J represents the comprehensive optimization objective function; η denotes the energy conversion efficiency; Y_{H_2} stands for the hydrogen production rate per unit volume of the

reactor; W_1 is the weight coefficient of η in the optimization, reflecting its priority; W_2 is the weight coefficient of Y_{H_2} in the optimization, indicating its relative importance.

Objective function 1 describes the relationship between energy conversion efficiency and key design parameters:

$$\eta = f_1(D, L, x_{in}, P, T, m_{cat}) \tag{6}$$

Objective function 2 characterizes the correlation between hydrogen production rate per unit volume and core design variables:

$$Y_{H_2} = f_2(D, L, x_{in}, P, T, m_{cat}) \tag{7}$$

where D denotes the inner diameter of the reactor; L denotes the length of the reaction section; x_{in} represents the position coordinates of the feed inlet; P represents the reaction pressure; T represents the reaction temperature; m_{cat} represents the catalyst loading constraint conditions.

The operating pressure of the reactor must be within the range that balances reaction thermodynamics and process safety:

$$P \in [0.1, 1.0] \text{ MPa} \tag{8}$$

The reaction temperature is restricted to the interval that ensures sufficient reaction activity while avoiding catalyst deactivation:

$$T \in [473, 623] \text{ K} \tag{9}$$

The catalyst loading must meet the minimum requirement for reaction rate and the maximum limit of reactor volume:

$$m_{cat} \in [100, 1000] \text{ kg} \tag{10}$$

4.1.1. CFD-ML-Based Reactor Design

In complex reactions such as dry methane reforming, the research by Alotaibi et al. demonstrates an effective method for integrating CFD with machine learning for industrial-scale fluidized bed reactor scaling. They generated extensive data covering various operating conditions and reactor configurations through CFD simulations. This data was then used to train machine learning models to predict reactor performance metrics, such as hydrogen yield and methane conversion rate. The ML model was subsequently employed to rapidly explore the vast design space and identify optimal reactor dimensions, feed inlet positions, internal components, and more, significantly enhancing hydrogen production at industrial scales [43]. In optimizing the structure of the CH_4 tri-reforming membrane reactor, Nasrabadi et al. also employed sensitivity analysis and machine learning methods. By analyzing the impact of various membrane reactor structural parameters, such as membrane area, catalyst packing method, and temperature distribution, on hydrogen yield, the ML model was able to quickly identify the key design variables most sensitive to performance and provide optimization directions. This approach offered valuable insights for the design of efficient membrane reactors [87].

4.1.2. Structural Parameter Optimization with Integrated Optimization Algorithms

In addition to being combined with CFD, machine learning models can also serve as objective functions for optimization algorithms, directly used to search for optimal structural parameters. Wang et al. developed a new framework integrating chemical reaction

models, ensemble learning methods, and whale optimization algorithms to optimize the Adsorption-Enhanced Steam Methane Reforming reactor. Their ensemble learning model, such as the ANN-based predictive model, accurately predicted hydrogen yield under various operating and design conditions. The whale optimization algorithm then used this predictive model as the objective function and successfully identified the combination of reactor operation and structural parameters that maximized hydrogen yield [76]. Wang et al. developed a machine learning-based optimization framework for the large-scale optimization design of solar-assisted hydrogen production processes. Their ML model learned the complex coupling relationships between subsystems such as the solar collector field, reforming reactor, and heat exchange network, enabling collaborative optimization of the entire system's equipment size and structural layout. This approach achieved a dual improvement in system economics and efficiency [88].

Machine Learning, through CFD-based collaborative design or integrated optimization algorithms, has enabled the efficient optimization of reactor structures, balancing the compactness and performance requirements of marine equipment. This provides technological support for the miniaturization and efficiency enhancement of marine hydrogen production reactors. Reactor structure optimization ensures the "hardware foundation," while the multi-objective collaborative optimization of process parameters serves as the core of "software control," directly determining the efficiency, cost, and emission balance of marine hydrogen production. This integrated approach helps to optimize both the physical reactor design and the operational parameters, ensuring that the system operates at maximum efficiency while meeting stringent environmental standards.

4.2. Multi-Objective Collaborative Optimization of Process Parameters

Once the reactor structure is determined, the multi-objective collaborative optimization of process parameters becomes key. Marine hydrogen production must balance conflicting objectives such as hydrogen yield, CO_2 capture rate, energy consumption, catalyst lifespan, and cost. The combination of Machine Learning with multi-objective optimization algorithms provides an ideal solution to this challenge. ML models can predict the outcomes of different process configurations, while multi-objective optimization algorithms can simultaneously consider all the relevant factors, finding the optimal balance between efficiency, sustainability, and cost-effectiveness for marine hydrogen production systems. This integrated approach ensures that the system can achieve high performance while meeting stringent environmental and economic requirements.

4.2.1. Surrogate Models for Accelerating Multi-Objective Optimization

Traditional multi-objective optimization methods, when directly based on high-fidelity process simulations, can result in significant computational burdens. ML, by constructing surrogate models with extremely low computational costs, makes in-depth and comprehensive multi-objective optimization feasible. In developing an interpretable ML framework for dry methane reforming, Roh et al. created a high-precision model that not only has the potential for multi-objective optimization but can also simultaneously optimize multiple indicators, such as conversion rate, selectivity, and catalyst stability. This approach enables the efficient exploration of the trade-offs between conflicting objectives, facilitating the development of more optimized and reliable catalytic processes with reduced computational effort [36].

4.2.2. Integrated Technical-Economic-Environmental Optimization

In recent years, researchers have increasingly focused on integrating technical, economic, and environmental indicators into optimization frameworks. Byun et al.'s Machine Learning model for methanol steam reforming can predict hydrogen yield, carbon emis-

sions, and the cost per unit of hydrogen under different operating conditions. This model provides a Pareto optimal solution set for marine hydrogen production, supporting parameter optimization for various navigation requirements. By considering both the performance and the environmental and economic impacts, the model helps identify the most efficient and sustainable solutions for hydrogen production in maritime applications, ensuring that multiple objectives are balanced effectively [46]. Mojtahed et al. applied Machine Learning techniques to model the waste heat recovery process for green hydrogen production and conducted a techno-economic analysis. Their ML model enables the rapid assessment of the economic feasibility of different process routes. By optimizing system configurations, the model aims to maximize both energy efficiency and economic benefits. This approach provides a powerful tool for evaluating various hydrogen production strategies, helping to identify the most cost-effective and energy-efficient solutions for green hydrogen production [89].

ML efficiently achieves the multi-objective collaborative optimization of process parameters. This method is particularly well-suited to the dynamic demands of marine hydrogen production under varying navigation conditions, balancing technical performance, environmental impact, and economic feasibility. Once the optimal operating range is determined through process parameter optimization, real-time monitoring of reactor performance and fault warning systems become crucial for ensuring the stable operation of the marine hydrogen production system. These monitoring tools help detect deviations from optimal performance, enabling proactive maintenance and minimizing system downtime, which is critical for maintaining the reliability and efficiency of onboard hydrogen production systems.

4.3. Reactor Performance Prediction and Real-Time Monitoring

Marine hydrogen production systems must adapt to fluctuations in the marine environment. To ensure their long-term stable and efficient operation, accurate prediction of reactor performance and real-time monitoring are crucial. ML can leverage data-driven approaches to implement soft measurement of key indicators and provide fault warnings. By continuously analyzing operational data, ML models can detect performance deviations early, allowing for timely intervention and preventive maintenance. This capability enhances system reliability, ensuring the hydrogen production process remains optimized even under dynamic environmental conditions.

4.3.1. Soft Measurement and Dynamic Prediction of Key Parameters

Reactor internal parameters, such as catalyst activity distribution, local hydrogen concentration, and coke formation, are difficult to measure in real-time online. Machine learning models can predict these parameters using easily accessible operating parameters, such as temperature, pressure, flow rate, and inlet and outlet compositions. Mustafa Tan and Cem Emeksiz developed a hybrid estimation model based on deep learning and probabilistic pooling for fuel cell parameter estimation. This model can be transferred to reforming reactors to enable online estimation of the reactor's internal states [90]. ML has enabled the soft sensing of key internal reactor parameters and the prediction of catalyst deactivation, providing support for the real-time operation, maintenance, and optimization of maintenance strategies for marine hydrogen production systems, thereby enhancing system operational reliability.

4.3.2. Catalyst Deactivation Prediction and Maintenance Strategy Optimization

Catalyst deactivation is a core challenge in reforming hydrogen production. Kumbhat et al.'s ML model can predict the deactivation rate and service life of nickel-based catalysts based on operating conditions and catalyst characteristics. The model recommends

optimized operating windows to extend catalyst lifespan and reduce costs. By accurately forecasting catalyst degradation, the model helps in adjusting operational parameters to minimize wear and tear, ensuring more efficient and cost-effective hydrogen production over the long term. This approach supports the development of sustainable catalyst management strategies in reforming processes [91].

ML enables the soft measurement of key reactor parameters and the prediction of catalyst deactivation, providing crucial support for the real-time operation and maintenance optimization of marine hydrogen production systems, thereby enhancing operational reliability. Local performance monitoring provides the basis for single-point control, while system-level integration and intelligent control achieve the collaborative optimization of energy across the entire vessel, adapting to the dynamic changes in the marine environment. This comprehensive approach ensures that the hydrogen production system remains efficient, responsive, and sustainable under varying operational conditions at sea.

4.4. System Integration and Intelligent Control Strategies

Reactor performance monitoring provides the basis for local control, while the core requirement of marine hydrogen production is system-level integration and intelligent control. ML plays a key role in the collaboration between the hydrogen production system and the vessel's energy management, adapting to the dynamic changes in the marine environment. By optimizing both the hydrogen production process and energy distribution across the vessel, ML enables real-time adjustments, ensuring that the system remains efficient and responsive to environmental conditions, enhancing the overall performance and sustainability of marine hydrogen production.

4.4.1. Collaborative Optimization of Multi-Energy Systems

Clean energy vessels typically integrate energy sources such as solar and wind power, and ML optimizes the matching of hydrogen production and vessel load, improving overall energy efficiency. For example, Huang et al. used ML to optimize diesel auto-reforming conditions, providing stable hydrogen-rich syngas for the ship's Solid Oxide Fuel Cell [70]. Song et al. explored the ML application in sodium borohydride-catalyzed hydrolysis for hydrogen production. Their system integration approach is directly adaptable to marine scenarios—using ML to process complex multi-variable relationships, ensuring real-time synchronization of hydrogen production and the ship's power demand [42]. On a larger scale, Hong et al.'s multi-objective optimization framework integrates deck-mounted solar panels, dynamically adjusting hydrogen production rates to reduce reliance on traditional fuels [92]. These studies demonstrate the potential of ML in enabling low-carbon, economically efficient operation of multi-energy systems in marine vessels.

4.4.2. Intelligent Adaptive Control

The operating environment of ships is complex, involving load fluctuations, ocean climate changes, and other factors, which require hydrogen production systems to have adaptive control capabilities. Machine learning models can be used to develop real-time control strategies, enhancing system resilience. For example, ML can enable real-time monitoring and predictive maintenance, which can be applied to ship hydrogen production systems. By quickly detecting anomalies through sensor data, it helps prevent downtime. L.G.de Araujo et al. highlighted in their review that data-driven models provide the foundation for advanced control strategies, such as adjusting hydrogen production parameters to adapt to changes in ship speed [65]. In addition, "real-time data-driven decision-making" can be extended to ship scenarios, where machine learning algorithms optimize hydrogen storage and delivery to address the uncertainties of the marine environment [26].

It is evident that machine learning plays a crucial role in the integration and intelligent control of hydrogen energy systems in clean energy ships. By optimizing multi-energy collaboration, it reduces carbon emissions, and through adaptive control, it enhances system robustness, providing technical support for the transition of ships to sustainable energy.

ML enables the collaborative optimization and intelligent adaptive control of marine hydrogen production systems and multi-energy sources, effectively addressing fluctuations in the marine environment and load variations. This technology provides core support for the decarbonization and reliability of ship propulsion systems. By continuously adjusting hydrogen production and energy distribution in response to dynamic conditions, ML enhances system performance and ensures efficient, sustainable operations, contributing significantly to the low-carbon transformation of marine vessels.

ML has formed a complete application system in the design of marine hydrogen production equipment, process optimization, performance monitoring, and system integration. From equipment to systems, and from static design to dynamic control, ML technology fully adapts to the compactness, real-time requirements, and stability needs of marine scenarios. By integrating these aspects, ML provides a feasible engineering pathway for the decarbonization of the shipping industry, optimizing both hydrogen production and energy management to meet the operational challenges of marine vessels. This comprehensive approach ensures sustainable, efficient, and reliable systems, contributing to the industry's shift toward cleaner energy solutions.

5. Conclusions and Future Outlook

This paper systematically reviews the research progress of machine learning in fuel reforming hydrogen production technology, focusing on three core areas: catalyst research, reaction process modeling, and equipment and system optimization. Through a thorough literature review and analysis, it is clear that ML applications have evolved from atomic-scale catalyst design to micro-level reaction mechanism analysis, and ultimately to system-level collaborative optimization, forming a complete empowerment chain from the micro to the macro scale, and from basic research to applications.

In the field of catalysts, ML has significantly shortened the R&D cycle and reduced trial-and-error costs through high-throughput virtual screening, performance prediction, and synthesis path optimization, especially for nickel-based, precious metal, and bimetallic systems. In reaction process modeling, ML is deeply integrated with first-principles calculations and computational fluid dynamics, constructing high-precision kinetic models and reaction path analysis tools, enhancing the understanding and predictive capabilities of complex reforming reaction networks. In system optimization, intelligent methods such as reinforcement learning and genetic algorithms are used in reactor design, multi-parameter collaborative optimization, and energy efficiency management. These methods, particularly in integration with marine power systems, demonstrate their unique potential in real-time monitoring, fault diagnosis, and intelligent control.

However, current research still faces a series of challenges that hinder the transition of ML models from laboratory settings to real industrial applications, especially in maritime propulsion systems, which require unique environmental and safety standards.

First, data quality and scarcity are bottlenecks for model generalization. Many studies rely on limited experimental data or idealized simulation datasets, which fail to capture the complexity of real industrial scenarios, particularly the variable relationships under the dynamic conditions of ships. This leads to a decline in model performance when transferring across systems or scales. Models trained on literature datasets carry the risk of overfitting, as they may perfectly "memorize" data features under specific experimental conditions but lack true learning of underlying physical and chemical principles. This

results in model failure when dealing with new operating conditions or new catalyst systems. The "black-box" nature and lack of interpretability of models restrict their use in safety-critical marine energy systems. Pure data-driven models without physical principles are not transparent in their decision-making process, making them difficult to fully trust in the engineering community and hindering a deeper understanding of catalytic mechanisms or system failure modes.

Second, the unique challenges of the maritime application environment introduce additional difficulties. Firstly, dynamic and real-time requirements: fluctuations in ship load require the hydrogen production system to respond rapidly, placing demanding requirements on the online computation speed of complex ML models. Secondly, the harsh environmental adaptability: the high salinity, humidity, and corrosiveness of the marine environment require catalysts and reactor materials to have stronger resistance to interference, yet most current research focuses on idealized laboratory environments, lacking specialized datasets and evaluation frameworks for this particular environment. Thirdly, stringent safety standards and spatial constraints: limited space and high safety standards aboard ships require that ML models be deeply integrated with safety systems, with decision-making processes meeting audit requirements set by maritime regulators. Fourthly, the economic bottleneck of the entire supply chain: cost control across the entire green hydrogen supply chain—from renewable hydrogen production to port storage and transport to ship refueling—is critical. Current research often focuses on optimizing individual steps and lacks an integrated solution for cross-scale, multi-objective collaborative optimization from the perspectives of total cost of ownership (TCO) and carbon footprint.

Despite these significant challenges, the empowering value of ML is tangible and quantifiable. In hydrogen production process optimization, AI-driven real-time diagnostic and control systems analyze operational data to dynamically adjust key parameters, such as voltage, current, and temperature, in electrolyzers and other critical equipment. In water electrolysis hydrogen production technology, energy efficiency (based on higher heating value, HHV) in traditional systems typically ranges from 70% to 80%. According to published technological backgrounds, with advancements in technology, PEM electrolyzers are expected to achieve energy conversion efficiencies close to 90%.

In the field of catalyst development, machine learning accelerates traditional catalyst research and development cycles by using high-throughput virtual screening and performance prediction, while also reducing experimental trial-and-error costs. Additionally, in reactor optimization, ANN-based surrogate models have reduced the average simulation time of chemical reactors by approximately 93.8%, with no significant loss in accuracy, which corresponds to a speedup of nearly 15 times.

The integration of machine learning in ship-based fuel reforming hydrogen production technology shows transformative potential. However, to fully implement smart hydrogen technologies in shipping decarbonization, targeted research is still needed to address several open challenges. Based on the core research gaps identified in this paper, five specific and actionable research directions are proposed to move the field from theoretical exploration to engineering practice.

(1). Physics-guided Machine Learning Models and Generalization Improvement

The current purely data-driven models lack generalization ability under complex marine conditions, which is a key bottleneck for their application. Existing models often rely on idealized simulation data and fail to capture real-world scenarios such as high salinity, high humidity, and dynamic loads aboard ships. Future research should prioritize the development of physics-informed neural networks, incorporating thermodynamic laws and reaction kinetics as hard constraints during the model training process. By integrating the partial differential equations that describe heat and mass transfer in reform-

ers into the loss function, the model can ensure that the predictions align with physical laws. Additionally, combining interpretable tools like SHAP and PDV will enhance model transparency, allowing the predictions to be verified against engineering principles. This "physics + data" dual-driven approach will bridge the gap between black-box models and engineering reliability, significantly improving the model's extrapolation ability under extreme marine environments.

(2). Edge Computing-empowered Digital Twin Real-time Optimization System

Traditional computational fluid dynamics simulations take days, which does not meet the millisecond-level response requirements for ship-based hydrogen production systems. The solution lies in developing lightweight machine learning models, such as RF or ANN, that can process sensor data in real-time on edge devices and optimize reformer operating parameters. These models should be deeply integrated with a digital twin framework to create a full lifecycle simulation system that includes dynamic load responses. This hybrid architecture not only reduces computational overhead but also supports closed-loop control systems to adjust operational parameters in real-time, accommodating sudden fuel demand changes or environmental disturbances.

(3). Cross-scale Multi-objective Collaborative Optimization Paradigm

Current research often focuses on optimizing individual components, neglecting the coupling effects between molecular, device, and system-level scales. Future research should establish a cross-scale collaborative optimization system, considering catalyst composition design, reactor structure optimization, and ship energy efficiency management in parallel. At the system level, reinforcement learning agents can dynamically adjust reformer output to balance hydrogen production with IMO carbon intensity requirements.

(4). Maritime-specific Material Development and Safety Evaluation Framework

The highly corrosive marine environment poses challenges to the durability of catalysts and reactors. Future research should leverage ML to design corrosion-resistant materials. Additionally, a safety evaluation system that complies with maritime regulations should be developed. This system could train ML models based on historical failure data and real-time sensor inputs to predict equipment aging trends or hydrogen leakage risks, and integrate with the ship's safety management system to automatically isolate faulty equipment or activate emergency venting protocols.

(5). Green Hydrogen Supply Chain Intelligent Scheduling System

The economic bottleneck of green hydrogen arises from insufficient collaboration across the entire supply chain—from renewable hydrogen production to ship refueling. Future research should focus on developing reinforcement learning-driven scheduling systems to coordinate the operation of electrolyzers at offshore wind farms with hydrogen demand from ships. Furthermore, ML platforms could track the entire lifecycle carbon footprint of hydrogen energy to ensure compliance with international low-carbon certification standards. Such innovations are expected to reduce the leveled cost of hydrogen across the maritime supply chain, making it more economically competitive with traditional fuels.

The research topics in this field have rapidly shifted from early-stage catalyst performance prediction (around 2022 and earlier) to the integration of multiscale simulations and system-level intelligent control (since 2022). Geographically, North America, Europe, and East Asia are the leading research hubs, focusing on fundamental algorithms and catalyst discovery, system integration and policy analysis, and rapid industrial applications, respectively. However, cases of translating laboratory results into onboard practical applications remain limited, with most research still in the simulation or experimental phase. Future work must strengthen collaboration between academia, industry, and research institutions, promoting the validation of "AI + mechanism" integrated models in real ship environments, overcoming dynamic environment adaptability challenges, and constructing intelligent

hydrogen supply chains deeply integrated with ships, ports, and storage and transportation facilities. With advancements in algorithms and computational power, ML-enabled fuel reforming hydrogen production technology will undoubtedly make a significant contribution to achieving the International Maritime Organization's 2050 decarbonization targets for shipping, injecting new momentum into global maritime sustainability.

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Appendix A

Appendix A.1

Table A1. Research on Catalysts and Design Optimization Driven by Machine Learning for Hydrogen Production via Fuel Reforming (Full Dataset).

Hydrogen Production Methods	Machine Learning Approaches	Inputs	Outputs	Model Performance	References
Methane Dry Reforming	Machine Learning-Driven Optimization of Catalyst Design	Catalyst Materials (Metal–Organic Framework, MXene, Biochar-Based), Reaction Parameters	Conversion Rate, Selectivity, Carbon Deposition Resistance	ML can enhance the depth of data analysis, improve the accuracy of catalyst design, accelerate catalyst development, and guide experimental design	[93]
Methane Dry Reforming	Interpretable Machine Learning, utilizing SHapley Additive exPlanations and Partial Dependence Values tools	Catalyst Parameters (Type of Promoter, Support Properties), Reaction Conditions (Temperature, Fuel-to-Oxidant Ratio, Space Velocity)	Methane Conversion Rate, Carbon Dioxide Conversion Rate, Hydrogen Production Rate, Carbon Deposition Amount	The model exhibits high prediction accuracy, with experimental validation showing an R^2 value greater than 0.9 and a low RMSE	[36]
Methane Dry Reforming	Machine Learning for Unbiased Data Set Analysis	Catalyst Elemental Composition	Catalyst Activity and Carbon Deposition Suppression	The model demonstrates high prediction accuracy and plays a key role in identifying elements such as aluminum and niobium	[52]

Table A1. *Cont.*

Hydrogen Production Methods	Machine Learning Approaches	Inputs	Outputs	Model Performance	References
Methane Dry Reforming	An explainable CatBoost model, enhanced with interpretability tools, to improve transparency	Reaction temperature, gas hourly space velocity (GHSV), calcination temperature, nickel loading, and catalyst structural parameters	Methane Conversion Rate	The CatBoost model predicts the methane conversion rate with an R^2 value of 0.91, and experimental validation shows an error of less than 10%	[37]
Methane Dry Reforming	Application of Machine Learning in CT Image Segmentation	Microstructural Data, Computed Tomography (CT), Energy Dispersive Spectroscopy (EDS)	Catalyst Degradation and Microcrack Formation	Phase distribution is achieved through CT segmentation and EDX; ML is used solely for image segmentation	[94]
Photocatalytic Hydrogen Production	Artificial Neural Network (ANN) Model	Catalyst Type, Light Conditions, Reaction Time	Hydrogen Production Efficiency (μmol g^{-1} h^{-1})	The ANN model predicts hydrogen production efficiency with high accuracy, as confirmed by experimental validation	[95]
Photocatalytic Hydrogen Production	Machine learning is used solely for optimizing parameters in biological hydrogen production (ANN)	Fermentation Temperature, pH Value, Substrate Concentration	Biological Hydrogen Production Rate	Prediction of Biological Hydrogen Production Rate and Optimization of Operating Parameters	[96]
Biomass Catalytic Pyrolysis for Hydrogen Production	Random Forest, Regression Model	Reaction Temperature, Carbon Content, Calcination Temperature, Nickel Loading, Biomass Type	Hydrogen Production	The RF model predicts hydrogen production with an R^2 of 0.78 and an RMSE of 0.47	[27]
Catalytic Steam Reforming of Biomass Tar	Machine Learning Algorithms (RF, ANN) for Optimizing Nickel-Based Catalysts	Reaction Temperature, Catalyst Support, Additive Type, Nickel Loading, Calcination Temperature	Toluene Conversion Rate	The RF model for predicting toluene conversion rate was evaluated with an accuracy of 0.99 and an AUC of 0.92	[38]
Carbon Dioxide Methanation	Explainable Machine Learning (SHAP Analysis), XGBoost	Reaction temperature, nickel content, calcination temperature, particle size, reduction time	Carbon dioxide conversion rate, methane selectivity	The XGBoost model demonstrates high prediction accuracy, as verified through experimental validation	[39]
Dry Reforming of Methane (CO_2 Methanation)	Multilayer Perceptron and Nonlinear Autoregressive Exogenous (NARX) Neural Networks	Calcination temperature, reduction temperature, reaction temperature, reaction time, and Ni loading	Methane conversion rate, carbon dioxide conversion rate	The NARX neural network achieves the highest R^2 = 0.998 and the lowest MSE = 3.24 $\times 10^{-9}$	[40]

Table A1. *Cont.*

Hydrogen Production Methods	Machine Learning Approaches	Inputs	Outputs	Model Performance	References
Hydrogenation of carbon dioxide to produce methanol	Machine learning models (ANN, Support Vector Machine Regression)	Catalyst composition and reaction conditions (temperature, pressure, space velocity)	Carbon dioxide conversion, methanol selectivity, carbon monoxide selectivity, and methanol space-time yield	The ANN demonstrates high predictive accuracy for all four output variables, with $R^2 > 0.9$	[58]
Hydrogen Production through the Catalytic Hydrolysis of Sodium Borohydride	Machine Learning–Assisted Optimization (RF, AdaBoost, and GBDT)	Catalyst Synthesis Parameters (Composition of Co–P–B/ZIF-67) and Reaction Conditions (Temperature and Concentration)	Hydrogen Yield and Reaction Rate	The R^2 values of the Random Forest model range from 0.956 to 0.995. After optimization, the removal of outliers enhanced the prediction stability	[42]
The Water–Gas Shift reaction (WGS)	Bayesian optimization integrated with RF and ANN	Catalyst Composition and Operating Conditions (Temperature, Feed Composition, Contact Time, Calcination Time)	Catalyst Activity, Stability, and Cost-effectiveness	Improvement in Optimized Catalyst Performance Indicators and Prediction Accuracy ($R^2 > 0.9$)	[97]
Sorption-Enhanced Chemical Looping Reforming	Artificial Neural Network	Operating Parameters (Temperature, Pressure, Flow Rate) and Catalyst/Adsorbent Properties	Methane Conversion Rate, Hydrogen Purity, and Carbon Dioxide Removal Efficiency	The ANN model demonstrates high prediction accuracy, $R^2 \geq 0.9889$	[41]

Appendix A.2

Table A2. Application of Machine Learning in Equipment Design and System Process Optimization (Full Dataset).

Application Scenarios	Machine Learning Approaches	Inputs	Outputs	Model Performance	References
Steam Methane Reforming	Deep Neural Networks Combined with Random Search Optimization Algorithm	Reactor length, feed flow rate, heat flux, S/C (steam-to-carbon ratio)	Hydrogen yield	The DNN model offers high prediction accuracy and the capability for multi-objective optimization	[28]
Electrically Heated Steam Methane Reforming	Model Predictive Control (MPC) based on Recurrent Neural Networks (RNN) and Improved Long Short-Term Memory (LSTM) Networks	Current, feed flow rate, argon flow rate, reactor temperature	Hydrogen yield	The LSTM-MPC under disturbances results in an error of less than 3%, with tracking accuracy significantly outperforming traditional control methods	[78]

Table A2. *Cont.*

Application Scenarios	Machine Learning Approaches	Inputs	Outputs	Model Performance	References
Integration of SMR and NET Power Cycle	ML Model Prediction and GA-based Process Optimization	Reaction temperature, pressure	Levelized Cost of Hydrogen (LCOH), carbon dioxide emissions, system efficiency	After optimization, the LCOH decreased to \$3.39 kg^{-1}, and the energy required for capture was reduced by 54%	[77]
Methane Autothermal Reforming	Deep Reinforcement Learning combined with Random Forest models and Q-learning algorithm	Oxygen-to-methane ratio, steam-to-methane ratio, temperature	Operating expenses, carbon emissions per unit time, hydrogen yield	After DRL optimization, OPEX decreased by 10%, carbon footprint was significantly reduced compared to traditional processes, and hydrogen yield increased by 13%	[79]
Solar-driven Methanol Steam Reforming	Grey Relational Analysis (GRA) and Genetic Algorithm Optimized BP Neural Network (GA-BPNN)	Reaction temperature, methanol flow rate	Methanol conversion rate, hydrogen yield, carbon monoxide selectivity	The GA-BPNN model demonstrates high prediction accuracy, outperforming traditional prediction models	[84]
Solar Photo-voltaic/Thermal (PV/T) System for Hydrogen Production	Stacked Ensemble Model (Random Forest, XGBoost)	Solar irradiance, temperature, water flow rate, PV/T type, operating conditions	Hydrogen yield	The R^2 value of the stacked model for prediction accuracy is 0.9986	[98]
Solar-Driven Green Hydrogen Production	Fast Fourier Transform (FFT) and Singular Spectrum Analysis (SSA) combined with Deep Learning (GRU)	Global Horizontal Irradiance (GHI)	GHI multi-step prediction, photovoltaic power generation estimation, hydrogen yield prediction	The R^2 value of the noise-robust model for Global Horizontal Irradiance prediction is 0.99, which supports the achievement of low-emission targets	[99]
Fuel Cell and Ammonia-Hydrogen Internal Combustion Engine Hybrid System	Integrated Deep Learning (PCA, MCNN, SVM)	Voltage, current, temperature, pressure, fault signals	Fault diagnosis accuracy, system reliability	Overall diagnostic accuracy of 98.15%, with multi-system fault diagnosis at 96.67%	[85]
Solid Oxide Fuel Cell and Integrated System	Multiphysics model combined with Deep Learning and Multi-objective Genetic Algorithm	S/C, operating temperature, fuel flow rate	Power density, maximum temperature gradient, carbon deposition rate	Achieving a multi-objective balance of "significantly reducing carbon deposition, maintaining high power density, and controlling safe temperature gradients"	[100]
Biomass-driven SOFC Combined Heat and Power (CHP) System	Triple-objective optimization using the Grey Wolf Algorithm	Biomass flow rate, operating temperature, pressure, fuel utilization efficiency	Power output, hydrogen yield, ammonia yield, system cost	After optimization, cost is minimized, efficiency is maximized, and ammonia yield is maximized	[101]

Table A2. *Cont.*

Application Scenarios	Machine Learning Approaches	Inputs	Outputs	Model Performance	References
Biomass Gasification for Hydrogen Production	Gradient Boosting Regression Model combined with Particle Swarm Optimization	Biomass type, temperature, reaction time, biomass concentration, pressure, reactor type	Hydrogen yield	After PSO, the test R^2 of the Gradient Boosting Regression model is 0.958, and its cross-validation R^2 is 0.917	[86]
Co-gasification of biomass and plastic for hydrogen production	Attention Mechanism-based Multi-Layer Perceptron Model (agMLP)	Temperature, plastic percentage, HDPE particle size, RSS particle size	Hydrogen yield	The agMLP model predicts hydrogen concentration with an R^2 of 0.997, robustness to be improve	[102]
Co-gasification of Biomass and Plastic	Gradient Boosting	Particle size, temperature, mixing ratio	Hydrogen yield	The gradient boosting model exhibits the best prediction performance, with an R^2 of 0.99	[103]
Industrial-Scale Vacuum Pressure Swing Adsorption (VPSA)	ANN combined with evolutionary algorithm optimization	Feed flow rate, purge ratio, feed pressure, vacuum pressure	Hydrogen purity, recovery rate, generation rate, energy consumption	After ANN optimization, purity reached 99.99%, and the efficiency was 45.2%	[104]
Pressure Swing Adsorption (PSA)	ANN and NSGA-II Optimization	Feed composition, pressure sequence, temperature, adsorbent properties	Hydrogen purity, recovery rate, production rate	The ANN model exhibits high prediction accuracy, while NSGA-II identifies the optimal operating conditions	[105]
Pressure Swing Adsorption Hydrogen Purification	DNN and NSGA-II Optimization	Adsorbent sequence, CuBTC bed length, feed flow rate, adsorption pressure	Hydrogen purity, recovery rate	The prediction accuracy of DNN yields an R^2 of 0.98, and NSGA-II identifies the optimal solution	[106]
Proton Exchange Membrane Water Electrolyzer (PEMWE)	Cascaded Feedforward Neural Network (CFNN)	Current density, temperature, anode material, water flow rate	Cell potential	The CFNN model predicts the cell potential with an R^2 of 0.99998, demonstrating high stability	[107]
Supercritical Water Gasification (SCWG)	SVR, ABR, DT, RF, GBR	Temperature, concentration, catalyst, residence time	Hydrogen yield, gas composition	The GBR model predicts hydrogen yield with an R^2 of 0.997, and the MSE for H_2 prediction using GBR is 0.54	[108]
Supercritical Water Gasification for Hydrogen Production	Integrated Tree AdaBoost Regressor (ELA) combined with Differential Evolution Optimization (DEO)	Biomass type, reaction temperature, residence time, catalyst concentration	Hydrogen yield	The ELA model predicts hydrogen yield with an R^2 of 0.95 and an RMSE of 0.091	[109]

Table A2. *Cont.*

Application Scenarios	Machine Learning Approaches	Inputs	Outputs	Model Performance	References
Dehydrogenation of Liquid Organic Hydrogen Carriers (LOHC)	Machine Learning and Genetic Algorithm Integrated Framework	Temperature, pressure, catalyst composition, support type	Methylcyclohexane conversion rate, toluene selectivity	The conversion exceeds 90%, and the selectivity exceeds 85%. For predictive performance, the R^2 value for methylcyclohexane (MCH) conversion prediction is 0.962, while the R^2 value for toluene selectivity prediction is 0.991	[35]
Hydrogen Compressed Natural Gas (HCNG) Engine Waste Heat Recovery	Stepwise Linear Regression (SLR) Model	Exhaust gas temperature, S/C, pressure, thermal load	Hydrogen yield	The SLR model predicts hydrogen production with an R^2 of 0.99, with the minimum RMSE of 0.074 and minimum MAE of 0.06	[110]
Biogas Dry Reforming Process	Random Forest Classification Model combined with SHAP	Temperature, pressure, operating time, humidity	Operating condition classification, fault prediction	Random Forest state prediction accuracy, with SHAP identifying key variables	[111]
Hybrid Wind-Hydrogen Energy Plant	Multi-Agent Reinforcement Learning (MARL)	Wind speed, electricity price, grid status, air density	Day-ahead trading profit, hydrogen yield, grid balance revenue	The MARL strategy increases total profit by 4%, with an annual increase of 7 million euros	[112]
Hybrid Renewable Energy System	Mountain Gazelle Optimizer and Transformer Architecture (MGO-Transformer)	Wind speed, temperature, relative humidity, cloud cover	Direct Normal Irradiance (DNI) prediction, energy cost, hydrogen cost	The MGO-Transformer predicts DNI with an R^2 of 0.998, and the system's LCOH is \$5.26 kg^{-1}, making it more economical	[113]

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